

**Amrita Vishwa Vidyapeetham**  
**Amrita School of Computing, Amritapuri**  
**BTech Computer Science and Engineering (AI)**

22AIE311 Software Engineering (Project based)

**LAB SHEET 1**

| <b>CO</b> | <b>Course Outcomes</b>                                                                                            |
|-----------|-------------------------------------------------------------------------------------------------------------------|
| CO01      | Understand the basic principles of Software Engineering.                                                          |
| CO02      | Understand how to choose the appropriate SDLC models depending on the user requirements.                          |
| CO03      | Understand the concept of Requirements Engineering.                                                               |
| CO04      | Understand the concept of Software Design.                                                                        |
| CO05      | Understand how to apply the knowledge, techniques, and skills in the development of software and its maintenance. |

**Group Members:**

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***Requirement Engineering – Inception and Elicitation Phase***

***Objective***

To understand and perform the Inception and Requirement Elicitation activities of Requirement Engineering for a given software project.

## **PART A – INCEPTION PHASE**

### **Task 1: Problem Identification**

[CO03]

1. Define the problem clearly.

In scanned documents (in particular those in multilingual areas, such as in India), sometimes the text on one page has multiscribed/multilingual text (e.g., English + Malayalam + Tamil + Hindi + Urdu). Currently available commercial and free-open-source OCR systems typically need individual language selection, cannot handle mixed-script pages, cannot easily identify individual languages where available on the page, and do not give a structured format of output that can be searched, tagged, and sorted as to be useful with Indian official records. This results in inaccessibility, manual transcription effort, information retrieval errors, and lower productivity at the public administration, courts, universities, and archives.

2. What existing issues does the system solve?

- The user does not need to select what language to detect, as it is done automatically.
- Can detect with documents with mixed languages, as the AI can automatically distinguish the different languages.
- Legacy scanned documents can be searched easily, to get the required information.
- Provides structured, language-labelled output for further processing or integration.
- Can improve the efficiency of the people who handle many documents daily.
- Offers text tagging and documentation for better organization and future reference.

### **Task 2: Identify Stakeholders**

| Sl. No | Stakeholder Name                  | Role                                                                      | Interest in System                                                                                                                                             |
|--------|-----------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | End Users (Document Consumers)    | Government officers, administrative staffs, archivists, re-searchers etc. | Primary Users – They want easy upload, accurate extraction of data, fast searching and tagging, and reliable access to the content without technical knowledge |
| 2      | System Administrators             | IT/admin staff in departments/offices                                     | Manage users, monitor performance, ensure uptime, handle backups and scaling                                                                                   |
| 3      | Project Supervisors / Faculty     | Academic evaluators (For Colleges and schools)                            | Assess the quality of the documents and ensuring the document objectives are met.                                                                              |
| 4      | Government Departments / Archives | Owners of source documents                                                | Benefit indirectly by improved digitization, searchability, and preservation of records                                                                        |
| 5      | Legal and Compliance Officer      | Ensure data privacy and auditability                                      | Secure storage, access control, and non-sharing of user documents.                                                                                             |

### **Task 3: Define System Scope**

#### **1. What the system will do**

Accept scanned PDFs/images -> auto-identify multiple languages/scripts on a page -> preprocess images -> extract text per region with language tags -> store structured data, provide full text search, provide responsive web interface to view (side-by-side original and text), search (keywords + language filter), highlight, tag documents/regions, copy/export text, user authentication and document management.

#### **2. What the system will NOT do**

- Live OCR of images captured with a camera (only uploaded scanned files).
- Translating languages automatically.
- Direct connectivity with physical scanners.
- Beyond fundamental tagging Advanced document classification or entity extraction.
- Documents can be shared publicly (per-user only, privately).
- Native mobile application (only web responsive).

#### **3. System boundaries**

In: File upload (PDF/image), user authentication (JWT), OCR pipeline with custom AI), postgres storage, elastic indexing, react UI, docker/k8s deployment.

Out: Physical documents scanning equipment, third-party translation services, shared document storage, mobile app, high-tech analytics/dashboard.

## **Task 4: Identify Constraints**

### **1. Budget constraints**

Constrained to student/academic project budget; can only use open-source tools; could use free-tier cloud credits on training/deployment of Gpus.

### **2. Time constraints**

The Course 22AIE311: This course requires 4-5 months to be accomplished within the semester; model training, integration, testing and documentation will have to be completed within the semester.

### **3. Technical constraints**

Custom AI OCR model should be trained/optimized (cannot work with Tesseract only); the inference latency should be achieved; Kubernetes on student-accessible resources

### **4. Legal constraints: The law requires data privacy (user documents isolated, cross-user access not allowed)**

## **Task 5: Feasibility Study**

### **1. Technical Feasibility:**

Feasible. Primary technologies (Django, React, PostgreSQL, Elasticsearch, OpenCV, PyTorch/TensorFlow) are not new and have a great following. Finetuning on datasets of Indian scripts is possible to produce custom multilingual OCR.

### **2. Economic Feasibility:**

Highly feasible. The tools are all open source and free. Very low costs, only cloud GPU (training) or cloud Kubernetes credits which can be paid with university resources or free tiers.

### **3. Operational Feasibility:**

Practical and very advantageous. Basic skills with browsers are required of end users (government staff, researchers). The system saves a lot of manual effort and enhances accessibility where multilingual settings are involved hence good reception.

### **4. Schedule Feasibility:**

Feasible within the timeline of the semester.

**Task 6: Select Elicitation Techniques**

**Interview** - This is selected since one will be able to know the real problems that exist in the current OCR tools, what is wanted (e.g., language support, the need of tags) and what expectations regarding the usability of OCR products are likely to be in the multilingual environment.

**Prototyping (UI/UX)** - Chosen to test the UI/UX idea (e.g., side-by-side viewer, highlighting, language color-coding) early, get early feedback on the workflow, and re-define what is required before full implementation of visual-intensive features, in particular.

**Task 7: Prepare Interview Questions (Minimum 8 Questions)**

- What kinds of documents you scan do you most regularly interact with, and what are the languages/scripts that you see on the same page?
- What are the most significant issues that arise when attempting to search or any other information in multilingual scanned PDFs/images?
- What is the significance of the system being able to automatically recognize and process various languages so that you do not have to choose them?
- Would you have rather a scanned image with extracted text displayed side by side with the original scanned image? What visual representations (e.g. color-coding per language) would you appreciate?
- How often do you have to label or annotate certain sections of documents to refer or share with other members of your team in future?
- What type of search abilities do you require (e.g. keyword search, language filter, phrase search)?
- What should the acceptable waiting time be to process a one-page document or 10 pages PDF?
- Do you have any particular security or privacy issues that you consider when uploading official/government documents to a web system?

## **Task 8: Identify Requirements:**

Classify the requirements into,

### **1. Functional Requirements**

- User registration, login, session management based on the JWT (FR-AUTH-01 to 03).
- Adding a document (PDF/image, multi-page, size verification) (FR-DOC-01)
- View status, List, delete documents (FR-DOC-02,03)
- Multi-language per-page automatic bounding boxes detection (FR-OCR-01).
- Per-language/region text extraction with ordering preserved (FR-OCR-02)
- Image preprocessing (noise removal, deskew, etc.) (FR-OCR-03)
- Structured JSON output per document (FR-OCR-04)
- Full-text keyword search with snippets (FR-SEARCH-01)
- Language-filtered search (FR-SEARCH-02)
- Keyword highlighting in viewer (FR-SEARCH-03)
- Document and segment tagging/annotation (FR-TAG-01,02)
- Side-by-side viewer with color-coded regions (FR-VIEW-01)
- Copy text / export as txt or JSON (FR-VIEW-02,03)

### **2. Non-Functional Requirements (System quality attributes)**

- Performance: OCR <10s/page, search <1s, UI load <2s
- Security: HTTPS, hashed passwords, row-level security, JWT signed tokens
- Scalability: Horizontal scaling via Kubernetes HPA
- Usability: Responsive design, clear error messages
- Availability: 99% uptime target, health checks, backups
- Reliability: Liveness/readiness probes, async processing (Celery)

**Task 9: Prepare Use Case List**

| Use Case ID | Use Case Name                | Actor         |
|-------------|------------------------------|---------------|
| UC-01       | Register Account             | End User      |
| UC-02       | Login / Logout               | End User      |
| UC-03       | Upload Scanned Document      | End User      |
| UC-04       | View Processing Status       | End User      |
| UC-05       | View Extracted Text          | End User      |
| UC-06       | Search Documents by Keyword  | End User      |
| UC-07       | Filter Search by Language    | End User      |
| UC-08       | Highlight Keywords in Viewer | End User      |
| UC-09       | Assign Tags to Document      | End User      |
| UC-10       | Annotate Text Region         | End User      |
| UC-11       | Copy / Export Extracted Text | End User      |
| UC-12       | Delete Document              | End User      |
| UC-13       | Manage User Accounts         | Administrator |
| UC-14       | Monitor System Health        | Administrator |

